

Maru Metall – Project guide

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We are building an AI quoting assistant that reads the project files Maru already collects on its shared drive, prices them with Maru's own rate file, and lets the estimator approve every number. The pitch focuses on three values for Maru: a better estimate of material and carbon footprint, more accurate quotes, and enough time saved to bid on more projects.

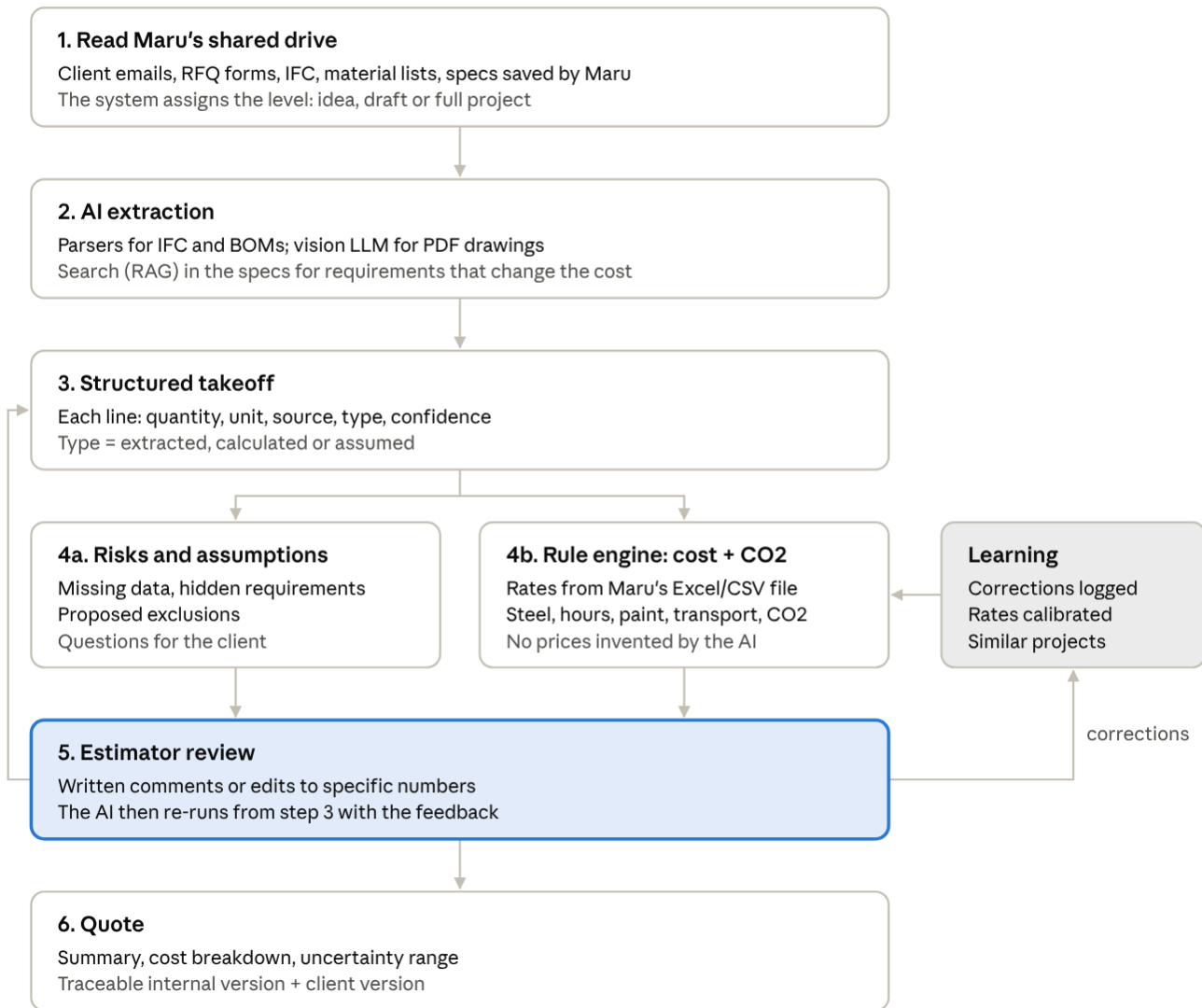
1. The solution

The system fits into how Maru already works; nobody has to change channel.

1. **Input stays as today.** Clients send inquiries by email, tender portals or meetings. Maru saves everything in the project folder on its shared drive ("01 Input for pricing" in the example data).
2. **AI reads the drive.** It picks up new folders, reads emails, RFQ forms, IFC models, material lists and specifications, and extracts the features that drive the price: tonnes per category, plate share, paint area, corrosivity and fire class, scope, delivery.
3. **Rules calculate the price.** A rule engine multiplies those features by the rates in an Excel/CSV file that Maru owns and updates (steel prices, hours norms, paint rates, transport, margins). The AI never invents a price.
4. **The estimator stays in control.** They see the draft quote, click any number to see where it came from, correct it, and approve. Corrections make the next quote better.

AI reads the drive, rules calculate, the estimator approves

PoC pipeline, from incoming files to the quote



solution architecture · 6 stages + learning loop

The same rule engine can compute an environmental footprint per quote, using a second column of factors in the same file (section 5).

2. What drives a quote

Maru's four example projects (148–776 t, €1.9–2.9 per kg) show that the price depends on a small set of features. The AI's job is to find them in the client files; Maru's rate file turns them into euros.

Feature group	What matters	Where the AI finds it
Commercial	Client, delivery address, bid deadline, offer validity, currency, warranty	Emails, RFQ forms
Scope	What is included or excluded: modelling, erection, fasteners, fire paint, gratings, handrails; phases and options	Emails, RFQ forms, colour-coded model screenshots
Quantities	Tonnes per category (columns, beams, trusses, bracing, plates), plate share, paint area	IFC / Tekla models, material lists
Technical specs	Execution class, steel grade, recycled content, corrosivity class, fire rating	Specifications
Logistics	Destination, number and type of trucks, longest piece	Drawings, delivery address
Strategic context	Workshop load, client relationship, past offers in €/t	Maru's own records

Maru's rate file holds the prices the rules apply: steel prices with quote date, waste factor (1.05–1.10), labour rate and time norms, paint rates by corrosivity class (9–25 €/m²), galvanising, packaging, transport per truck and margins. Maru uses a quick €/t method per category and a detailed part-by-part method; the system should offer both.

What the examples teach us → Sort of AI training on the examples

- **Late spec changes are expensive.** In the DEPO Tallinn tender, R30 fire paint was added mid-tender (+€280k) and the quote went through 5 revisions. Catching such requirements on day one is real value.
- **Product mix moves the price.** The Kotka skybridge is 62% plates, so it has lower productivity and the highest price (€2.87/kg).
- **Inputs are messy.** Mixed units, items repeated across IFC files, prices in NOK: the AI must flag these, not hide them.
- **Clients already ask for green steel.** The Akkasæter project required 80% recycled steel.

3. Data flow

Three roles, each doing what it is best at:

- **AI:** reads the drive, sorts the files, extracts the features above, flags risks and missing data, and turns the estimator's comments into proposed changes.
- **Rule engine:** calculates cost (and CO2) from the features and Maru's rate file. Same input, same result, every time.
- **Estimator:** reviews, corrects and approves. Nothing leaves without their sign-off.

Traceability. Every number in the draft shows where it came from (file, page or model element) and whether it was read from a document, calculated by a rule, or assumed because data was missing. Assumptions are listed so they can be confirmed with the client or written as exclusions.

Feedback loop. The estimator reviews the draft in one of two ways: a written comment in plain language ("client confirmed galvanising instead of paint, and the stairs are out of scope") or a direct edit of specific numbers (quantity 14 → 16). After the review, the AI re-runs the process from step 3: it rebuilds the takeoff with the feedback, the rules recalculate cost and CO2, and a new draft goes back to the estimator.

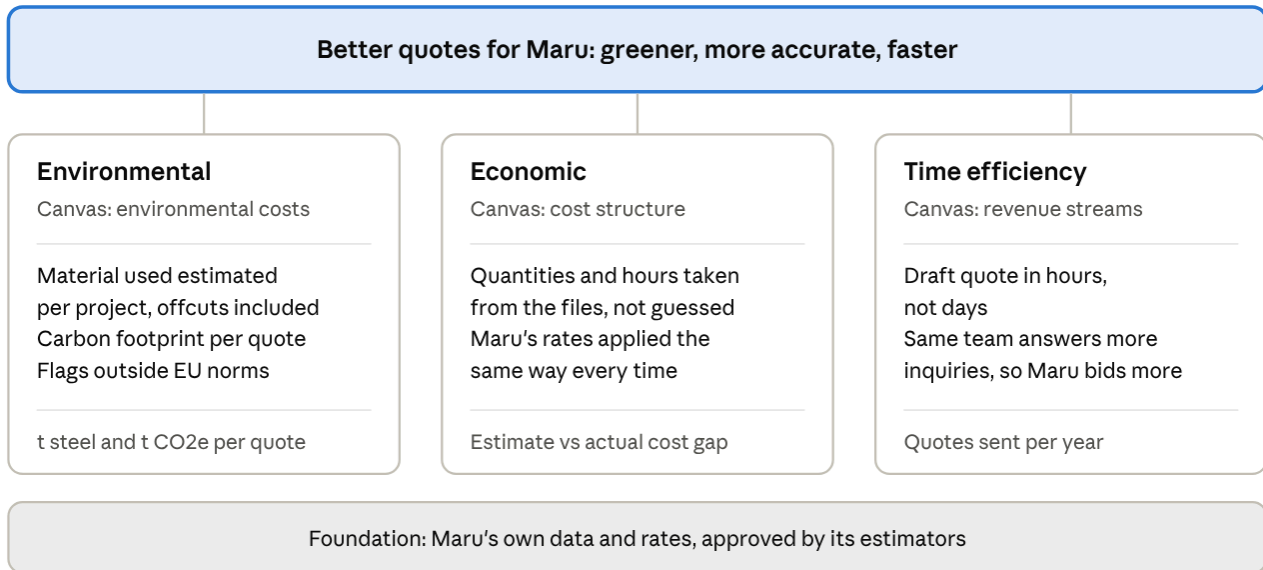
Over many quotes, the corrections calibrate Maru's rates, become examples that guide the AI on similar projects, and turn approved quotes into test cases.

AI will update its guidelines when a project is approved based on the Human feedback he got.

4. Value proposition

We focus the pitch on three values, each tied to one block of Maru's business model canvas and one number to show.

Three values carry the pitch



value proposition · 3 pillars

Value	Canvas block	What changes for Maru	Number to show
Environmental	Environmental costs and benefits (marked "minimum" today)	A better estimate of the material actually used, including offcuts, and a carbon footprint in every quote, with flags when values fall outside European norms	Tonnes of steel and tonnes of CO2e per quote
Economic	Cost structure (steel 30-50%, production 50-70%)	Quantities and hours come from the files and Maru's rates, so quotes are neither too low (lost margin) nor too high (lost bids)	Gap between estimate and actual cost
Time efficiency	Revenue streams (steel structures 84% of sales)	Drafts in hours instead of days, so the same team answers more of the 500+ yearly inquiries and wins more work	Quotes sent per year

5. Environmental module

Every quote gets two environmental outputs next to the price: a better estimate of the material actually used and a carbon footprint. Both follow the same pattern as pricing (AI extracts, rules calculate), and each number shows where its data came from. A flag warns when a value falls outside European norms.

Material used

- **Net steel.** Tonnes per profile and plate, taken from the IFC models and material lists, the same quantities used for the price.
- **Offcuts.** Today Maru's cost workbooks add a flat waste factor of 1.05–1.10 to the material, i.e. 5–10% extra steel bought. That is about 7–15 t on the 148 t Akkasæter project and 39–78 t on the 776 t DEPO tender. The system replaces the flat factor with a per-project estimate by profile and plate thickness (plate-heavy jobs waste more), and checks whether new parts can be cut from Maru's remnant stock before buying new steel.
- **Reclaimed steel.** Where the client's specification allows it, the AI flags standard members (beams, columns) that could come from reclaimed sections, and offers them as an option in the quote.

Carbon footprint

- **CO2 estimate:** steel tonnes × factor by production route, plus painted or galvanised area × factor, plus trucks × distance × factor. As an order of magnitude, blast-furnace steel is around 2 t CO2e per tonne and scrap-based electric-arc steel is well under 1 t; the real values should come from the suppliers' environmental product declarations (EPDs).

Where the data comes from

Data	Source	Status
Steel tonnes, plate share, paint area	IFC models and material lists in the "01 Input for pricing" folders	In the example data
Waste factor 1.05–1.10	Material lines of Maru's cost workbooks in the "02 Cost calculation" folders	In the example data; a flat assumption today

Data	Source	Status
Steel actually used	Final part lists in "04 Fabrication docs" and Maru's purchase records	Part lists in the example data; purchases to request
Remnant stock	Maru's warehouse list	To request (may not exist yet)
CO2 factors for steel, paint, galvanising	Suppliers' EPDs (EN 15804)	To request from Maru's suppliers
Transport emissions	Truck count and distance from the quote × a standard factor per km	Trucks in the workbooks; factor from a public database
Order-of-magnitude CO2 values	General industry figures	For the pitch only, replaced by EPD data

Flag: values outside European norms

The rules check each quote and show a flag next to the number concerned. The estimator decides what to do with it.

Flag	Raised when	Reference
Missing product data	Steel, paint or galvanising has no EPD following EN 15804	EN 15804, the European standard for construction product EPDs
No carbon declaration	A product lacks the global warming potential (GWP) data the revised EU Construction Products Regulation phases in from 2026	Revised CPR
Carbon above limit	The project's footprint exceeds the limit of the destination country or the client's tender	National rules, e.g. Finland's Building Act (2025) sets carbon footprint thresholds per building type
Waste above benchmark	Estimated offcuts exceed Maru's own average	Maru's past projects

Which rules apply depends on the destination country (Estonia, Finland, Norway, Sweden), so the list should be confirmed with Maru.

Why it matters for Maru

- Most of Maru's construction sales go to Scandinavia, where clients already set recycled content requirements; carbon data in the quote helps Maru win those tenders.
- Reusing offcuts lowers steel purchases, the biggest share of Maru's costs.
- The "environmental" boxes of the canvas, marked "minimum" today, become a measured benefit Maru can report.

To build it we need from Maru: a list of remnant stock (if they keep one), supplier EPDs or CO2 values, and how often clients ask for CO2 data.

6. The 3-minute pitch

Spend about a third of the time on how it works and two thirds on what it changes for Maru. Use one concrete number per part.

Time	Part	What to say
0:00-0:20	Hook	500+ inquiries a year, some taking hundreds of hours, all depending on a few experienced estimators
0:20-0:45	Problem	Messy inputs and late changes: in the DEPO tender R30 fire paint arrived mid-tender (+€280k) and the quote went through 5 revisions
0:45-1:30	Solution	Drive → AI extracts features → Maru's rate file prices them → estimator reviews → AI re-runs; show one number traced back to its page
1:30-1:55	Environmental	Material and CO2 estimated in every quote, with EU-norm flags; offcuts alone are 5-10% of the steel bought
1:55-2:20	Economic	Quantities from the files and Maru's own rates: fewer underpriced jobs, fewer lost bids
2:20-2:45	Time efficiency	Drafts in hours, so the same team bids on more of the 500+ inquiries
2:45-3:00	Ask	A proof of concept on one past project, with clear go/no-go criteria

Business case to fill with Maru's numbers: hours saved per year = 500 inquiries × average hours per quote × share of time saved. Add the value of extra bids: extra quotes sent × win rate × average order value.

7. Plan and next steps

The plan has four steps: understand Maru's data, build a small proof of concept on one past project, compare it with the real quote, then pitch.

- ☐ Confirm the challenge deadline, pitch format and evaluation criteria
- ☐ Split roles: business case and pitch, AI extraction, rules and demo
- ☐ Ask Maru for the missing pieces: time-norm tables, average hours per quote, win rate, remnant stock list, supplier CO2 data
- ☐ Turn Maru's cost workbook into a clean rate file (Excel/CSV) the rules can read
- ☐ Run the proof of concept on one example (Akkasæter is a good candidate: structured RFQ, IFC, material lists, recycled-steel requirement)
- ☐ Compare our estimate with Maru's real offer and note the gaps
- ☐ Build the canvas slide and the business case numbers
- ☐ Rehearse the 3-minute pitch with timing